

Mantavya Maan

B.S. Mathematics | Indian Institute of Technology Bombay
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Education

Indian Institute of Technology Bombay CPI: 8.86/10
B.S. in Mathematics; Minor in Game Theory and Logic, Department of Computer Science and Engineering Expected May 2027

Scholastic Achievements

- Secured top **0.49** percentile in **JEE Main 2023** examination out of **1.1 million** eligible candidates all over India.
- Among the top **0.97** percentile in **JEE Advanced 2023** examination out of **0.18 million** candidates all over India.

Professional Experience

Palpx | Summer Intern, Mathematical Modeling & Artificial Intelligence May 2026 – Jul 2026

- Formulated **4 predictive scoring models** via **statistical inference**, probabilistic modeling, and mathematical optimization.
- Applied **lambda calculus**, computational logic, functional modeling, and **Retrieval-Augmented Generation** to **AI systems**.
- Architected an **LLM orchestration platform** supporting **100+ AI models** using Bayesian inference, utility scoring, and routing.

Gradpipe | Machine Learning & Data Engineering Intern Nov 2025 – Dec 2025

- Architected a **high-dimensional talent discovery engine** using **vector databases** and hybrid retrieval for semantic matching.
- Engineered an **AST-based analysis pipeline** evaluating complexity, concurrency, and memory utilization for repository scoring.
- Designed an **LLM-as-a-Judge** evaluation framework using standardized benchmarks, reproducible scoring, and model validation.

Fino Payments Bank | Summer Intern, Data Analytics Jun 2025 – Jul 2025

- Analyzed **1M+ financial records** using **SQL**, **Pandas**, and **NumPy** to identify market trends and evolving customer behavior.
- Developed interactive **Tableau dashboards** tracking **20+ KPIs** to accelerate executive decision-making and business reporting.
- Collaborated across cross-functional teams to strengthen **data integrity**, reporting pipelines, and business analytics processes.

Projects

Neural Gateway | AI Infrastructure & Multi-LLM Orchestration May 2026 – Jul 2026

- Architected an **AI orchestration platform** routing **100+ LLMs** using Bayesian inference, Thompson sampling, and planning.
- Designed a **KNN vector-embedding parser** using FastEmbed, ONNX, semantic retrieval, caching, and sub-**10 ms** inference.
- Engineered a **FastAPI control plane** with OpenRouter synchronization, dynamic model registry, and adaptive policy enforcement.
- Developed **multi-stage execution planning** supporting verifier selection, fallback routing, response streaming, and audit trails.

DocuMind AI | Hybrid RAG Framework Jun 2026 – Jul 2026

- Built a **hybrid RAG framework** using **Qwen2.5:7B**, LangChain, Ollama, ChromaDB, Pinecone, and vector embeddings.
- Developed **multi-modal retrieval pipelines** supporting PDF, DOCX, PPTX parsing, semantic search, and grounded generation.
- Architected a **dual-mode inference system** enabling local execution, cloud deployment, runtime switching, and response streaming.

Lead Score Model | Mathematical Modeling and Decision Intelligence May 2026 – Jun 2026

- Built a mathematical lead qualification framework spanning over **10 scoring parameters** and **125+ CRM business variables**.
- Applied **Bayesian inference**, Beta confidence intervals, nonlinear optimization, and probabilistic modeling for robust lead ranking.
- Designed explainable lead prioritization using **sigmoid transforms**, exponential decay, adaptive score calibration, and optimization.

Independent Study | Riemann Surfaces and Algebraic Curves May 2026 – Jun 2026

- Studied **Riemann surfaces**, complex manifolds, and **algebraic curves**, establishing rigorous foundations in complex geometry.
- Analyzed **holomorphic mappings**, complex atlases, topological genus, and affine plane curves through rigorous formal proofs.
- Connected **Riemann surface theory** with **algebraic geometry** using meromorphic forms, function fields, and conformal maps.

Technical Skills

Languages & Frameworks: Python, C, C++, Java, SQL, JavaScript, FastAPI, Streamlit, LangChain, Ollama, LLM Orchestration
AI Infrastructure: LLMs, RAG, Prompt Engineering, Semantic Retrieval, Vector Databases, ChromaDB, Pinecone, Vector Search
Libraries & Tools: NumPy, Pandas, SciPy, scikit-learn, Matplotlib, Git, GitHub, Jupyter Notebook, Tableau, OOP, SQLite, L^AT_EX
Mathematical Concepts: Bayesian Inference, Thompson Sampling, Statistical Inference, Nonlinear Optimization, Semantic Parsing

Leadership & Activities

- Won **silver** in **Powerlifting** at the General Sports Championship, IITB and **bronze** at the Overall Institute Powerlifting Opens.
- Coordinated **15+ T20 matches** during Aahvan and served as Hostel **Secretary**, representing over **1200+ hostel residents**.
- Completed Summer of Science in **Data Structures & Algorithms** and Season of Code in **Competitive Programming**.
- Served as **Services** Coordinator at Mood Indigo, **Operations** Coordinator at E-Cell, and **Infrastructure** Coordinator at Techfest.